

Forecasting Unemployment Dynamics Across OECD Countries: A Comparative Time Series and Policy Analysis

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1. Introduction

Unemployment represents a fundamental indicator of economic health and social well-being, reflecting both macroeconomic conditions and structural characteristics of labor markets. Understanding its evolution over time is essential for policymakers aiming to design effective and targeted interventions.

This project investigates unemployment dynamics across OECD countries using time series forecasting techniques. In particular, it compares the performance of three models—Naïve, ARIMA, and Prophet—across multiple dimensions, including age groups, gender, and country-level data. Beyond predictive accuracy, the study aims to extract meaningful insights into the structural behavior of unemployment and translate these findings into actionable policy recommendations. This dual focus ensures that the analysis is not only technically rigorous but also economically relevant.

2. Data Description

The dataset used in this study is derived from the OECD Labour Force Statistics database and consists of quarterly observations of employment and unemployment across multiple countries and demographic groups. The analysis focuses on the working-age population, typically defined as individuals aged 15 to 64, while also exploring finer age group disaggregations.

To ensure comparability, the unemployment rate is computed as the proportion of unemployed individuals relative to the total labor force, defined as the sum of employed and unemployed individuals. This transformation provides a standardized metric that captures labor market conditions across different populations and time periods.

The dataset allows for a multidimensional analysis, enabling comparisons across age groups, gender categories, and countries. Such granularity is particularly valuable for identifying

heterogeneity in unemployment dynamics and assessing whether predictive performance varies across different segments of the population.

3. Methodology

The empirical analysis employs three distinct forecasting approaches, each representing a different level of model complexity. The Naïve model serves as a benchmark by assuming that future values will replicate the most recent observation. Despite its simplicity, it often performs surprisingly well in persistent time series.

The ARIMA (AutoRegressive Integrated Moving Average) model introduces a more structured approach by capturing temporal dependencies, trends, and stochastic components in the data. It is particularly well-suited for economic time series that exhibit autocorrelation and gradual evolution over time.

In contrast, the Prophet model is designed to handle more complex patterns by decomposing the time series into trend, seasonality, and holiday effects. While it is widely used in business forecasting contexts, its effectiveness depends on the presence of strong seasonal or nonlinear components.

Model performance is evaluated using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), which provide complementary measures of predictive accuracy. The analysis is conducted separately for different demographic and geographic segments to assess the robustness and generalizability of each model.

4. Exploratory Data Analysis

The exploratory analysis reveals several important characteristics of unemployment dynamics. As illustrated in Figure 1, the unemployment rate exhibits clear responses to major macroeconomic shocks, such as the global financial crisis of 2008 and the COVID-19 pandemic. These events lead to sharp increases in unemployment, followed by gradual recovery phases, highlighting the cyclical nature of labor markets.

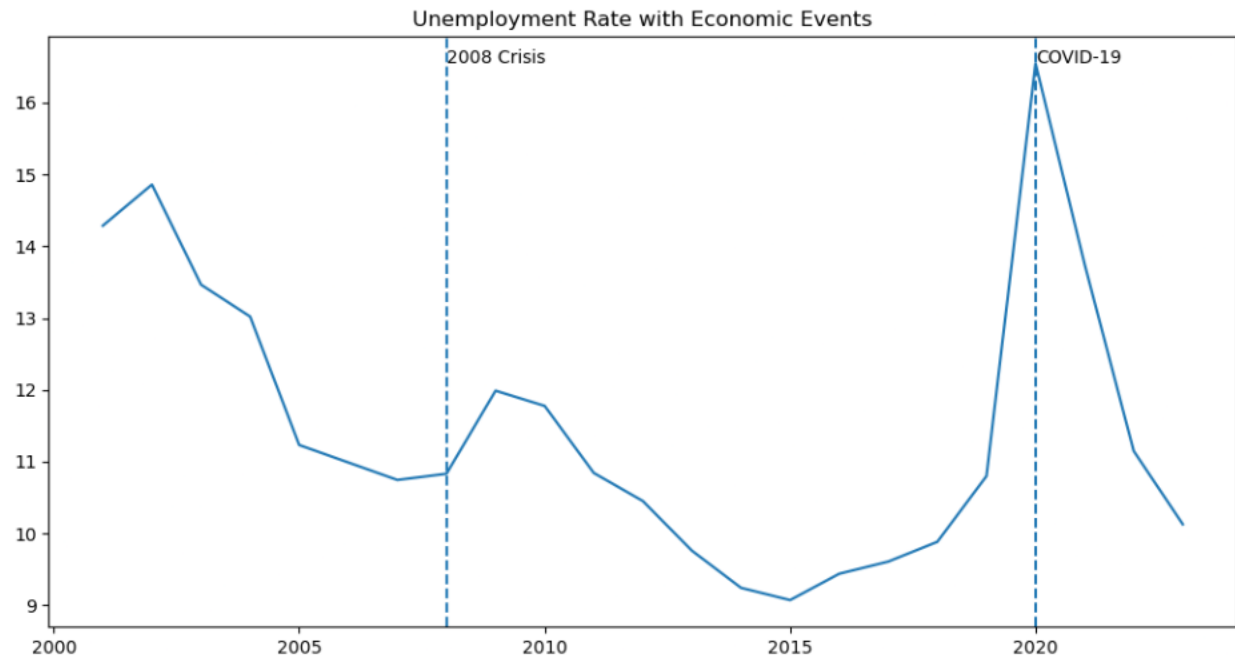


Figure 1: Time series of unemployment rate with 2008 crisis and COVID-19

A more granular analysis by age group (Figure 2) shows substantial heterogeneity in volatility. Younger individuals, particularly those aged 15 to 24, experience significantly higher fluctuations in unemployment rates compared to prime-age and older workers. This suggests that younger workers are more vulnerable to economic downturns and face greater uncertainty in labor market participation.

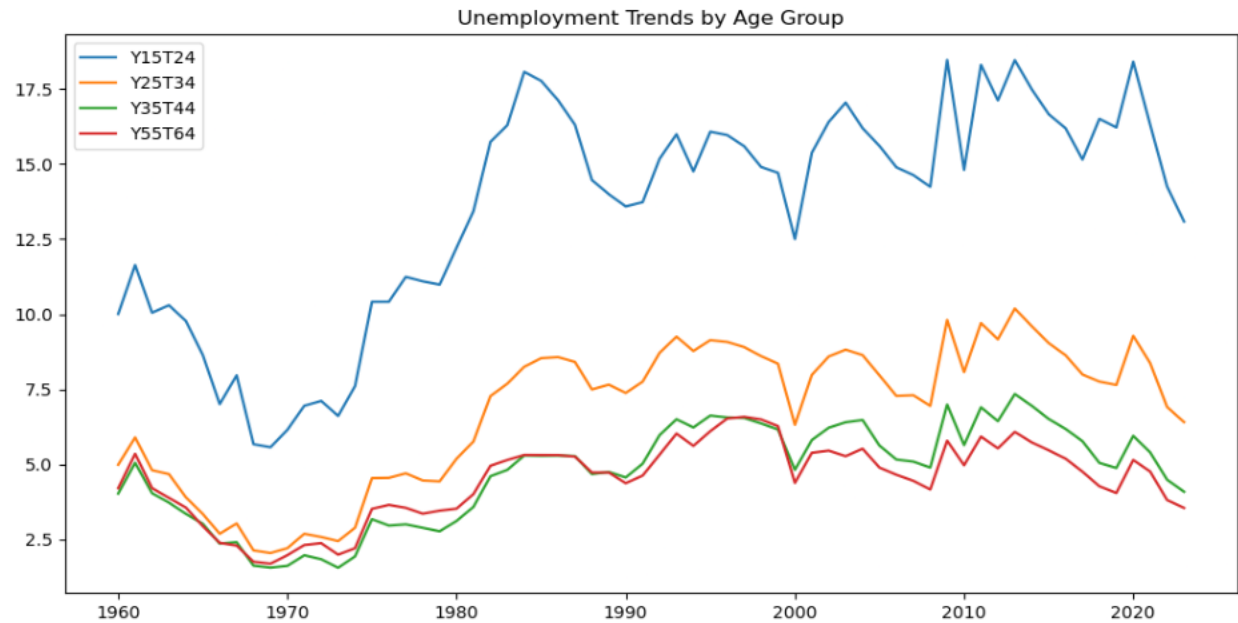


Figure 2: Multi-line plot of unemployment trends for selected age groups

In contrast, gender-based comparisons (Figure 3) indicate that unemployment trends for males and females are remarkably similar over time. While minor differences exist, they do not appear to be the primary drivers of variability in the data.

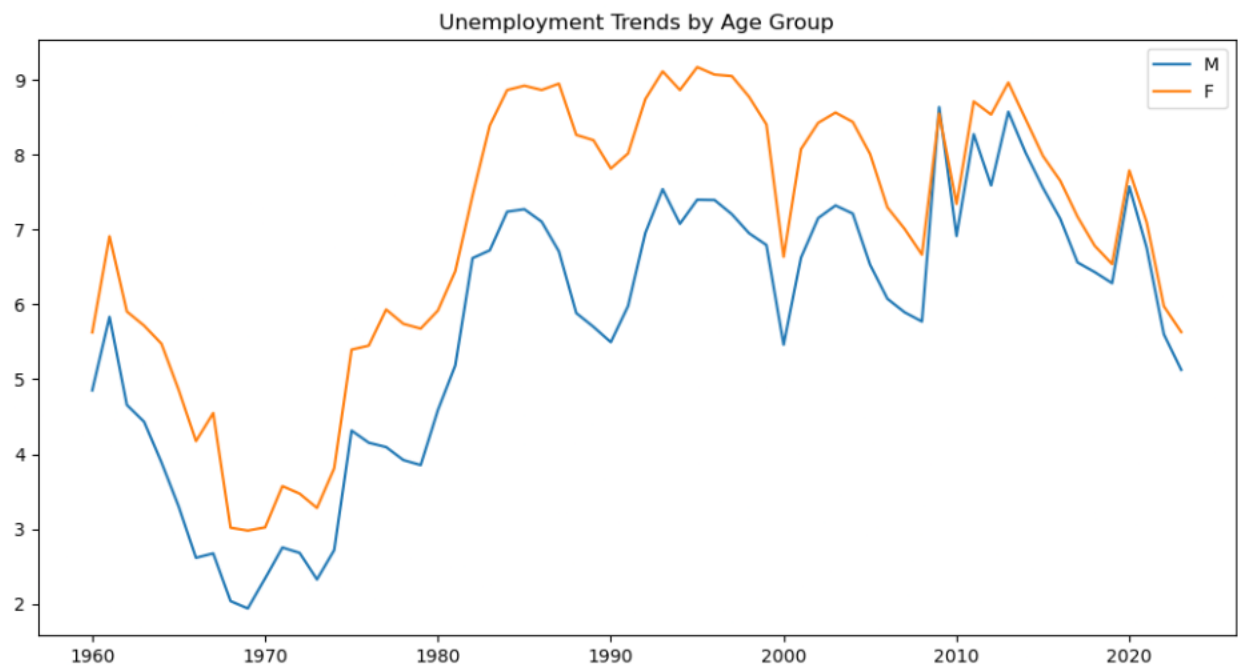


Figure 3: Line plot comparing unemployment rate by gender

At the country level (Figure 4), the analysis uncovers pronounced differences in unemployment dynamics. Some countries display relatively stable patterns, while others exhibit substantial volatility and sensitivity to economic shocks. This heterogeneity underscores the importance of institutional and structural factors in shaping labor market outcomes.

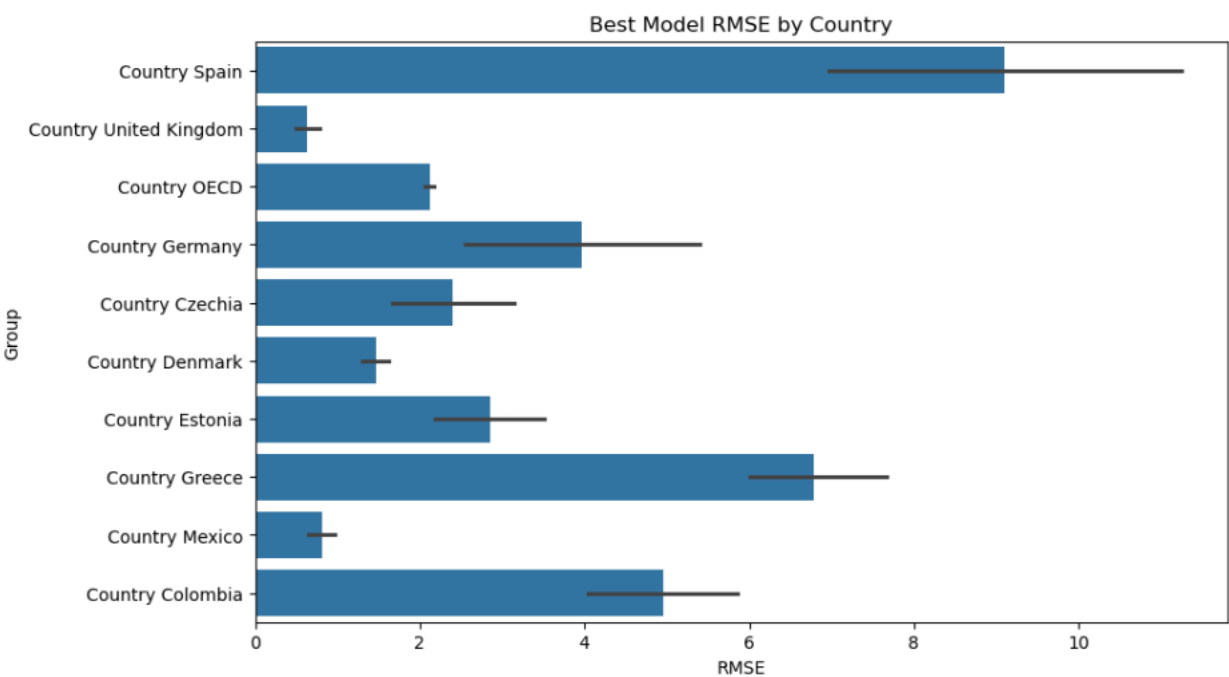


Figure 4: Selected Country comparison plot

5. Model Performance

The comparative evaluation of forecasting models reveals that simpler approaches often outperform more complex ones in this context. As shown in Figures 5 and 6, the Naïve and ARIMA models achieve similar levels of accuracy, with ARIMA providing only marginal improvements. In contrast, the Prophet model consistently underperforms, exhibiting significantly higher prediction errors.

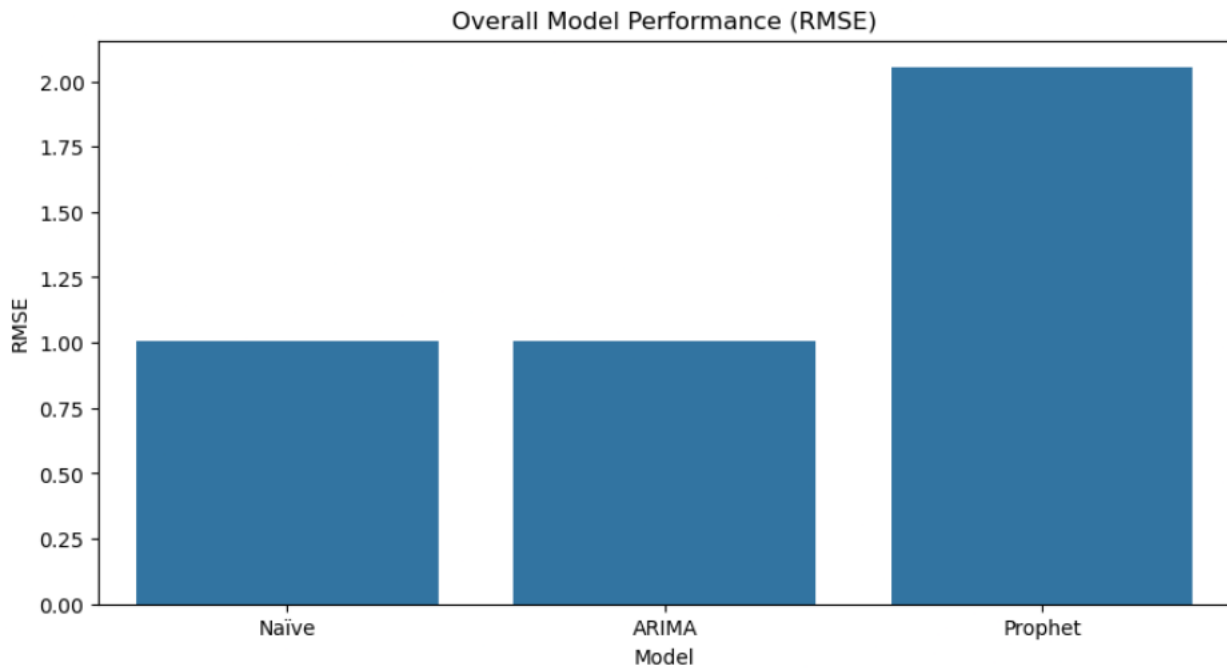


Figure 5: RMSE Barplot comparing Naive, Arima, Prophet

To better visualize performance differences across groups, a heatmap representation is used. This allows for a compact comparison of model errors across multiple segments simultaneously.

When disaggregating results by age group (Figure 7), it becomes evident that prediction errors are substantially higher for younger populations. This reflects the greater volatility and unpredictability of youth unemployment, which poses a challenge for all models considered.

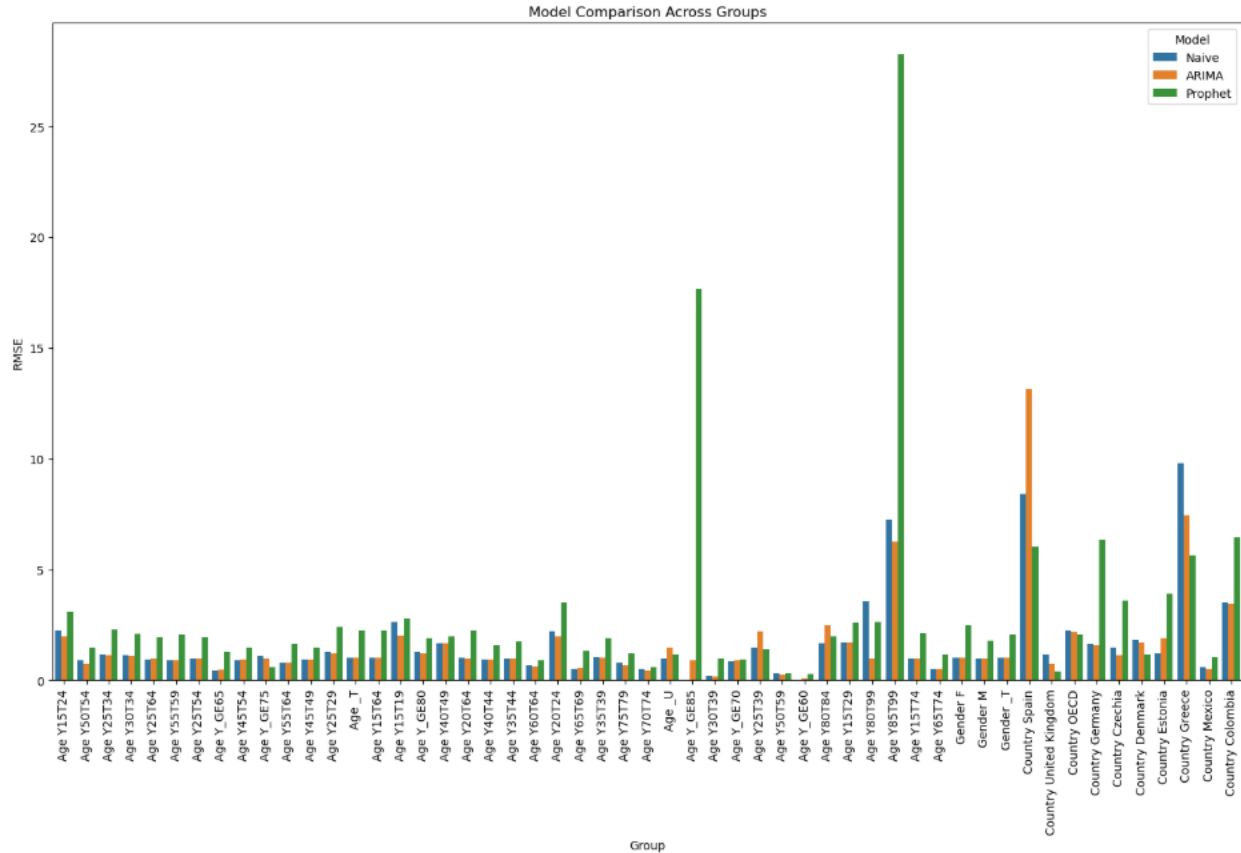


Figure 7: RMSE Barplot across age groups

Gender-based results confirm the findings from the exploratory analysis, with minimal differences in model performance between male and female populations. This suggests that gender does not introduce additional complexity from a predictive standpoint.

Country-level results (Figure 8) highlight strong variation in forecasting accuracy. Countries characterized by economic instability, such as Spain and Greece, exhibit higher prediction errors, whereas more stable economies, such as the United Kingdom and Denmark, are easier to forecast. These findings emphasize the role of macroeconomic stability in determining model performance.

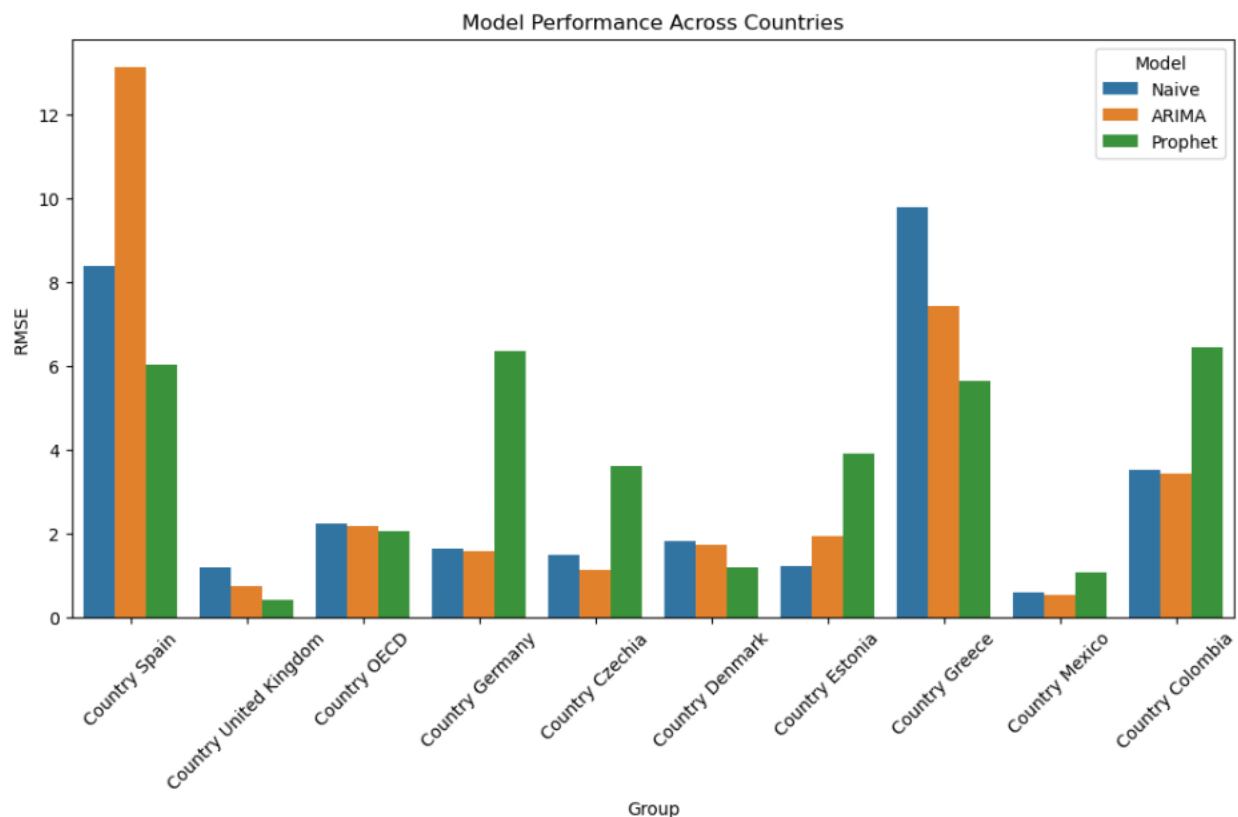


Figure 8: Model Performance Across Countries

6. Discussion

The results of this study provide important insights into the nature of unemployment dynamics. The strong performance of simple models suggests that unemployment follows relatively stable and persistent patterns, driven primarily by structural factors rather than complex seasonal fluctuations. This is further supported by the poor performance of the Prophet model, which relies heavily on the presence of strong seasonal components.

The forecast comparisons and residual analysis (Figures 9–11) indicate that while ARIMA captures general trends effectively, it struggles to fully account for sudden shocks and structural breaks. This limitation is particularly evident in younger age groups and in countries with volatile labor markets.

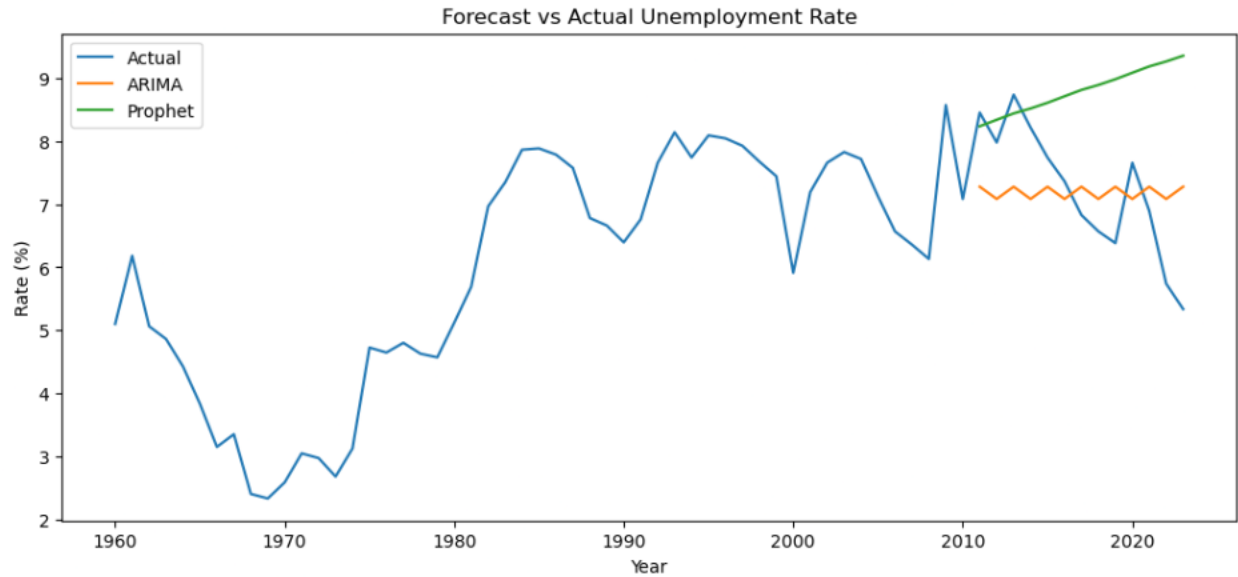


Figure 9: Forecast vs Actual

Overall, the findings suggest that unemployment dynamics are shaped by a combination of demographic characteristics and macroeconomic conditions. These factors introduce varying degrees of predictability across different segments, highlighting the importance of tailored analytical approaches.

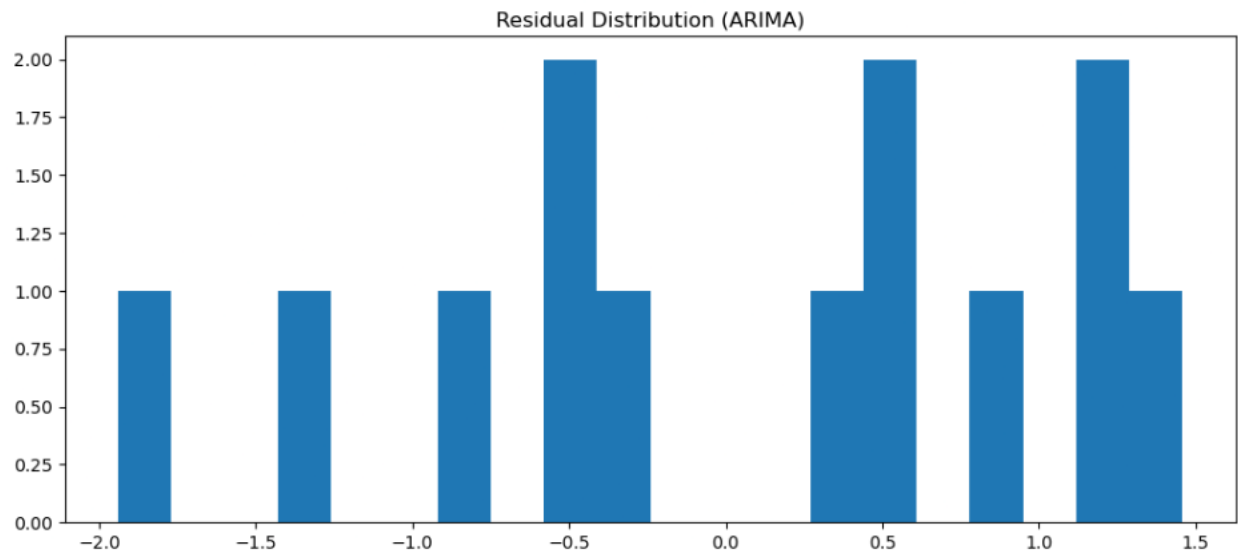


Figure 11: Histogram of Residuals

Additionally, the distribution of residuals provides insight into the presence of bias and variance in model predictions.

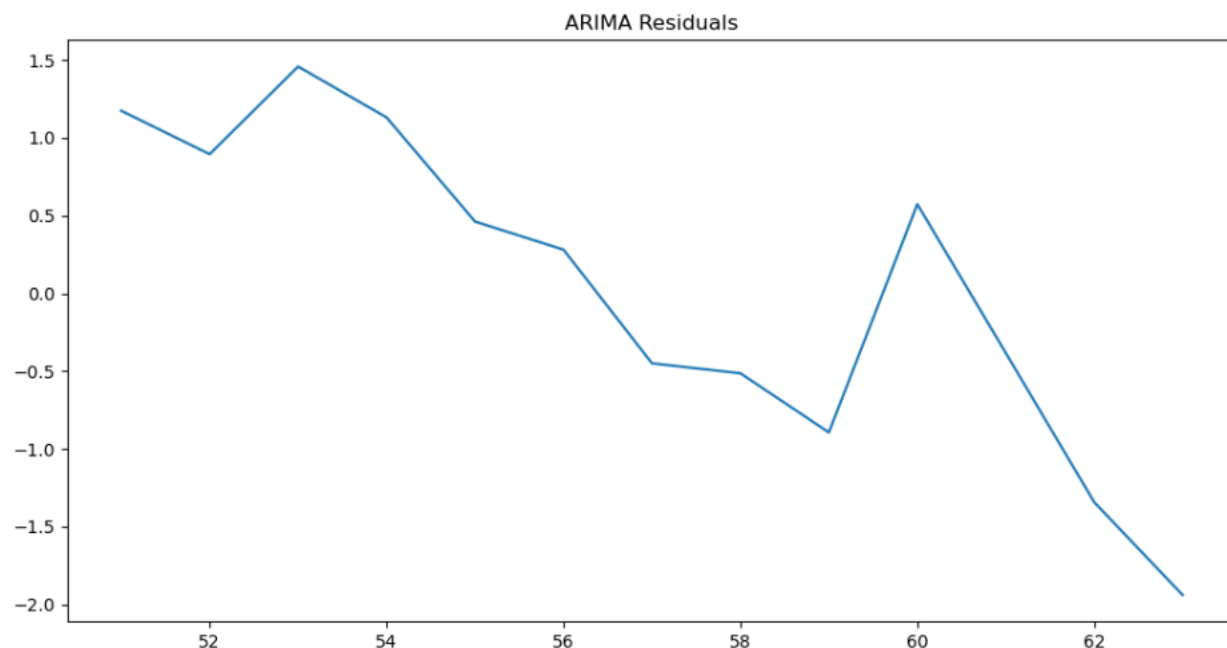


Figure 11: ARIMA Residuals

7. Policy Implications

The results of this analysis have significant implications for labor market policy. The high volatility and unpredictability of youth unemployment indicate a need for targeted interventions aimed at improving labor market entry. Policies such as vocational training programs, hiring incentives, and stronger links between education and industry can help mitigate the risks faced by younger workers.

For older populations, the relatively stable but less dynamic nature of unemployment suggests a different policy focus. In this case, initiatives aimed at reskilling and lifelong learning are essential to ensure continued labor market participation in the face of technological change.

The absence of substantial gender differences implies that broad labor market policies may be equally effective across genders. However, addressing structural barriers such as childcare availability and work-life balance remains important for improving overall participation rates.

At the country level, the observed heterogeneity underscores the need for context-specific policies. Economies characterized by high volatility require more robust social protection systems and flexible labor market institutions, while more stable countries can focus on long-term efficiency and adaptation to structural changes.

Importantly, the results suggest that policymakers should prioritize structural reforms over short-term reactive measures, as unemployment dynamics appear to be driven more by long-term factors than by short-term fluctuations.

8. Limitations

Despite its contributions, this study has several limitations. First, the analysis relies on a relatively limited set of variables, focusing primarily on employment and unemployment. The inclusion of additional macroeconomic indicators, such as GDP growth, inflation, or interest rates, could provide a more comprehensive understanding of labor market dynamics.

Second, the models used in this study are relatively basic and do not incorporate advanced machine learning techniques or hybrid approaches. While this choice allows for interpretability and comparability, it may limit predictive performance in more complex scenarios.

Third, the analysis does not explicitly account for structural breaks or regime changes, which can significantly affect unemployment dynamics, particularly during periods of economic crisis.

Finally, data quality and consistency across countries may introduce additional sources of variability that are not fully captured in the models.

9. Future Improvements

Several avenues for future research can enhance the robustness and scope of this analysis. One important direction is the integration of additional explanatory variables, such as macroeconomic indicators and policy variables, to improve model accuracy and interpretability.

Another potential improvement involves the use of more advanced forecasting techniques, including machine learning models such as Random Forests, Gradient Boosting, or Long Short-Term Memory (LSTM) networks. These approaches may be better suited to capturing nonlinear patterns and complex interactions in the data.

Furthermore, incorporating structural break detection methods could improve model performance during periods of economic disruption. This would allow the analysis to better account for sudden changes in labor market conditions.

Finally, extending the analysis to include scenario-based forecasting could provide valuable insights for policymakers by simulating the impact of different economic conditions and policy interventions.

10. Conclusion

This study provides a comprehensive analysis of unemployment dynamics across OECD countries, combining time series forecasting with policy-oriented insights. The results demonstrate that unemployment behavior varies significantly across demographic and geographic dimensions, with youth populations and economically volatile countries presenting the greatest challenges for prediction.

The findings highlight the effectiveness of simple forecasting models and emphasize the importance of structural factors in shaping unemployment dynamics. Most importantly, the analysis translates these insights into concrete policy recommendations, underscoring the need for targeted, context-specific, and long-term interventions.

By bridging the gap between predictive modeling and policy relevance, this project contributes to a deeper understanding of labor market dynamics and offers a foundation for more informed decision-making.

11. References

OECD. (2023). Labour Force Statistics: Employment and Unemployment

<https://data-explorer.oecd.org/>

The Gender Unemployment Gap. Stefania Albanesi & Ayşegül Şahin (2017). National Bureau of Economic Research (NBER) — NBER Working Paper 23743

Unemployment Rates by Selected Age Groups (OECD). OECD Employment Outlook (2016).

OECD iLibrary — [Table: Unemployment by age groups](#)

Gender Statistics – Unemployment (World Bank). World Bank / Gender Data. World Bank

Gender Statistics – Unemployment

https://erf.org.eg/app/uploads/2021/06/1624197845_384_1348394_movingbeyondunemploymentrateapeng.pdf?utm